

A Project Report on

**Client-Side Detection of Indirect Prompt Injection and
Unauthorized Data Exfiltration in Browser-to-AI Interactions
Using Multi-Indicator Content Analysis**

Submitted in partial fulfilment for award of degree of

Master of Technology

In Cybersecurity

Submitted by

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Under the Guidance of

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Candidate's Declaration

I, Sushanth Sridhar, hereby declare that I have completed the project work towards Master of Technology in Cybersecurity at REVA University on the topic entitled “Client-Side Detection of Indirect Prompt Injection and Unauthorized Data Exfiltration in Browser-to-AI Interactions Using Multi-Indicator Content Analysis” under the supervision of Sandeep, [Designation]. This report embodies the original work done by me in partial fulfilment of the requirements for the award of the degree for the academic year [Year].

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Name of the Student: Sushanth Sridhar

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Names of the Examiners

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Place: Bengaluru

Date:

Acknowledgment

I would like to express my sincere gratitude to my project guide, Sandeep, for the continuous guidance, technical feedback, and encouragement provided throughout this capstone project.

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List of Abbreviations

Abbreviation	Expansion
DOM	Document Object Model
TLS	Transport Layer Security
SNI	Server Name Indication
JA3 / JA4	TLS client fingerprinting methods (Salesforce JA3; FoxIO JA4)
MV3	Chrome Extension Manifest Version 3
CDP	Chrome DevTools Protocol
DLP	Data Loss Prevention
PII	Personally Identifiable Information
EDR	Endpoint Detection and Response
MITRE ATLAS	Adversarial Threat Landscape for Artificial- Intelligence Systems
OWASP	Open Worldwide Application Security Project
F1	Harmonic mean of precision and recall
TP / FP / TN / FN	True Positive / False Positive / True Negative / False Negative

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Abstract

(Not to exceed 1 page)

AI-powered browser agents (ChatGPT Atlas, Perplexity Comet, Claude in Chrome) now read and act on arbitrary web content on a user's behalf, and the open web has begun filling with content specifically crafted to hijack that behaviour — indirect prompt injection hidden in off-screen CSS, HTML comments, and metadata that a human never sees but an agent's context window does. Independent testing found ChatGPT Atlas blocked only 5.8% and Perplexity Comet only 7% of such pages, against 47–53% for conventional phishing defences in ordinary browsers, and Google recorded a 32% rise in malicious indirect-injection content between November 2025 and February 2026. Existing defences are vendor-side and reactive; a proactive, client-side sentinel that inspects page content before an agent ever reads it is largely unexplored.

This project builds and evaluates such a sentinel as a standalone system: a Chrome extension that scans the live DOM of any page for six independent indicators of hidden instructions (off-screen positioning, zero-width Unicode, suspicious HTML comments, hidden alt/ARIA text, JSON-LD metadata, and visible imperative language) and combines them with a noisy-OR multi-indicator score; a Go agent bridging the extension to a local Python scoring service and a Postgres store via Chrome's native messaging API; a real Zeek/Suricata network sensor performing JA3/JA4 TLS fingerprinting to identify known AI platforms and cluster unlisted “shadow AI” services; and an outbound data-loss-prevention gate that intercepts and classifies fetch/XHR bodies to known AI domains before they are sent, holding flagged content for user approval.

The system was evaluated end to end — not simulated — against a 70-page labelled synthetic dataset (30 benign, 10 deliberately weak “hard-negative” pages, 30 injected) visited by a 4-container test fleet, each container its own OS user and hostname, running the real extension, agent, and scoring pipeline. The multi-indicator detector (C) achieved F1 0.979 (precision 0.983, recall 0.975), against F1 0.911 for a visibility-only baseline (B) and F1 0.784 for a keyword-only baseline (A) — mirroring the same $C > B > A$ pattern from the author's prior capstone on network-based C2 detection. The result that most directly supports the multi-indicator design: on the hard-negative pages, the keyword-only baseline false-positived on 32 of 40 visits and the visibility-only baseline on 20 of 40, against 2 of 40 for the multi-indicator detector.

Building the system end to end surfaced nine real, documented defects and their fixes spanning every layer — a Chrome install-path assumption that silently broke content-script injection, a native-messaging registration gap, a TLS client-fingerprint (JA3) instability caused by Chrome's GREASE mechanism that silently defeated the shadow-AI clustering rule until re-keyed to JA4, and others — each caught by checking real output against expectation rather than trusting a clean-looking log, and each recorded with root cause and fix rather than smoothed over in the final narrative. The shadow-AI clustering rule was also confirmed to have genuine practical limits: once corrected, the same fingerprint that caught the two seeded “unknown AI” test domains also pulled in sixteen unrelated domains from the browser's own background traffic, demonstrating directly why the technique is a human triage signal rather than an automated verdict.

The findings support the conclusion that combining independent, cheaply-computed content indicators client-side detects indirect prompt injection substantially more reliably than any single indicator alone, while remaining fully inspectable — every flagged page shows exactly which indicators fired — and that this client-side vantage point is structurally unavailable to network-only or vendor-side defences.

Keywords: Prompt Injection, Indirect Prompt Injection, Browser AI Agents, Data Loss Prevention, Shadow AI, TLS Fingerprinting, JA3, JA4, MITRE ATLAS, Multi-Indicator Detection, Chrome Extension Security

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Plagiarism Report

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Paper Publications in a Journal/Conference Presented/White Paper

[None at time of submission]

Certificate for the Conference Presentation

[Not applicable]

Github Link

<https://github.com/Sushanth2624/browser-ai-sentinel>

Additional Reproduction Notes

[Content pending — see build plan for staging]